MoMa-Kitchen: A 100K+ Benchmark for Affordance-Grounded Last-Mile Navigation in Mobile Manipulation

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Abstract

Mobile manipulation commonly fails in the "last mile" of navigation: agents are rewarded for proximity to the target but not for whether the final base pose affords successful manipulation. We introduce MoMa-Kitchen, a large-scale benchmark with 127,343 episodes across 569 procedurally varied kitchen scenes that operationalizes last-mile navigation as dense, robot-conditioned floor affordance prediction. For each egocentric RGB-D view, MoMa-Kitchen provides aligned floor and global point clouds and a continuous affordance map indicating base poses that enable collision-free, kinematically feasible manipulation given the robot morphology. The benchmark defines a supervised task—predict the continuous floor affordance map from first-person RGB-D and robot parameters—and an evaluation protocol with RMSE, logMSE, Pearson correlation (PCC), cosine similarity (SIM), and a Manipulation Success Rate (MSR) metric linking predictions to end-to-end utility. We detail a fully automated data generation pipeline in BestMan[8] (PyBullet) using PartNet assets[4], with quality controls for visibility, collisions, and frame consistency. Evaluating representative point-cloud detectors [5, 6, 9] and a lightweight reference model (NavAff), MoMa-Kitchen reveals that affordance-grounded navigation is challenging yet diagnostic: NavAff attains RMSE 0.147, logMSE 0.0115, PCC 0.680, SIM 0.696 and improves MSR Top1/Top5 to 0.72/0.66 over adapted baselines. Ablations show the value of global context and visuomotor alignment, and qualitative sim-to-real results indicate plausibility in real kitchens from commodity RGB-D. MoMa-Kitchen provides a reproducible, robot-aware evaluation of last-mile navigation that complements existing embodied navigation benchmarks by measuring manipulability, not just proximity.

1. Introduction

Embodied navigation benchmarks have accelerated progress in goal-reaching and instruction following, often optimizing distance-based or success-at-goal metrics [1–3]. However, mobile manipulation introduces additional constraints—reachability, obstacle clearance, and workspace geometry—that determine whether a base pose actually affords the downstream manipulation. Conventional navigation rewards proximity, not manipulability; as a result, systems frequently fail at the last meter where base orientation and collision-free arm workspace are decisive.

We address this evaluation gap by introducing MoMa-Kitchen, a benchmark that measures "last-mile" navigation through floor-level, robot-conditioned affordance maps. Given an egocentric RGB-D observation and robot parameters, the task is to predict a dense map over nearby floor locations indicating the feasibility of executing a simple manipulation (e.g., grasp or suction) without collision. This focuses evaluation on what matters at execution time: selecting a base pose from which manipulation can succeed.

Design principles underlying MoMa-Kitchen include: (1) robot-awareness via explicit morphology conditioning; (2) realism and diversity through procedurally varied kitchens with rigid and articulated assets; (3) direct manipulability supervision via physics-based attempts; (4) contamination resistance through a fully automated, deterministic pipeline; and (5) diagnostic value through sliceable metrics and an MSR protocol that connects prediction quality to downstream success. Our main empirical findings show that (i) adapted 3D detection backbones [5, 6, 9] underperform a lightweight reference model tuned for affordances, (ii) global 3D context and visuomotor alignment significantly improve correlation with ground truth, and (iii) predictions transfer qualitatively to real kitchens with commodity RGB-D.

Contributions (1) **Benchmark:** MoMa-Kitchen, a 127k+ episode benchmark that evaluates last-mile navigation as continuous, robot-grounded affordance prediction in kitchens. (2) **Protocol & metrics:** a supervised task with RMSE, logMSE, PCC, SIM, and an MSR protocol linking predictions to manipulation outcomes. (3) **Automated pipeline:** a reproducible, simulation-based procedure for scene generation, manipulation trials, and dense affordance construction. (4) **Findings:** analysis across baselines, ablations, and qualitative sim-to-real transfer that characterize where current systems succeed and fail.

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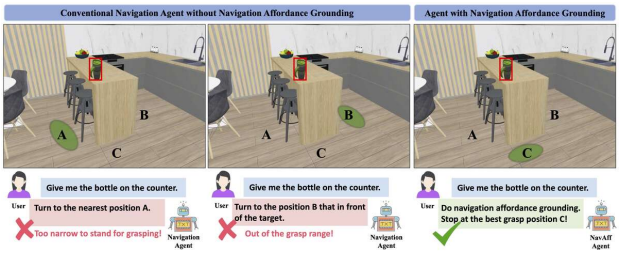

Figure 1. Problem motivation: proximity-based navigation can arrive near the target yet fail to afford manipulation due to reach and collision constraints. MoMa-Kitchen evaluates last-mile navigation via dense, robot-conditioned floor affordance maps.

2. Related Work

Embodied navigation and evaluation. Benchmarks in embodied navigation have emphasized photorealism, instruction following, and distance/success metrics [1–3, 7]. While crucial, these setups generally do not assess whether the final base pose supports manipulation, which is central in mobile manipulation.

Interactive and mobile manipulation benchmarks. Interactive navigation and embodied manipulation benchmarks consider contact and object interaction, e.g., Interactive Gibson and BEHAVIOR, and recent OVMM settings, yet most do not explicitly evaluate base-pose manipulability in a robot-conditioned manner. MoMa-Kitchen complements these by isolating last-mile base placement with physics-grounded supervision.

3D perception backbones. 3D point architectures such as PointNet++ [5], VoteNet [6], and H3DNet [9] provide strong geometric features and serve as adapted baselines in our study. Our reference model (NavAff) fuses global and floor-level 3D cues with robot morphology for dense affordances; it is provided to anchor the benchmark but is not the primary contribution.

Simulators and assets. We build on BestMan [8] (PyBullet) to procedurally generate kitchens and execute physics-based manipulation attempts, and leverage PartNet models [4] for diverse assets. This supports reproducibility and large-scale data generation.

3. Benchmark Design

Task. Given an egocentric RGB-D observation and robot parameters (e.g., base height and arm reach), predict a continuous floor affordance map over nearby base poses indicating the feasibility of executing a simple manipulation on a specified target from that pose. The supervised target is a dense feasibility score in $[0, 1]$ per floor point.

Scenes and targets. MoMa-Kitchen comprises 569 procedurally varied kitchen scenes with randomized layouts, clutter, and target placements, spanning 137 asset types (rigid, articulated, and obstacles). To induce reach/height variability, we sample six robot arms spanning consumer and industrial configurations. All assets are instantiated in physics simulation and undergo collision-aware checks.

Table 1. **Dataset Split Statistics.**

Split	Scenes	Configurations	Episodes
Train	456	$11,408$	$102,687$
Test (Unseen Scenes)	113	$2,747$	$24,656$

Sensing and alignment. Each episode records RGB, depth, a global point cloud, and a floor point cloud aligned to the robot base frame. We mount a first-person camera on the arm and select up to ten views per target that ensure line-of-sight.

Manipulation-grounded supervision. For each episode, we define a semicircular floor region parameterized by the arm reach, sample base poses, and execute grasp/suction attempts. Successes are transformed to the robot base frame and associated to nearest neighbors on the floor cloud, yielding sparse feasibility labels that are densified via k-NN Gaussian interpolation into continuous affordance maps.

Quality controls and splits. We reject views without line-of-sight, enforce collision-aware feasibility, and fix camera intrinsics and transforms for consistency. The pipeline is deterministic and fully automated. The dataset is partitioned into seen/unseen scenes; statistics are in Table 1.

Task variants. The evaluation scripts optionally support multi-view aggregation and multi-target episodes (reporting per-target metrics and macro-averages), enabling study of view selection and interactive replanning without altering the core labels.

4. Evaluation Protocol and Metrics

Supervised prediction. Models take as input the egocentric RGB-D and optional 3D point clouds and robot parameters, and output a dense scalar affordance per floor point.

Metrics. We report RMSE, logMSE, Pearson correlation coefficient (PCC), and cosine similarity (SIM) between predicted and ground-truth affordance values at annotated floor points. To assess utility, we compute **Manipulation Success Rate (MSR)** by selecting the Top-k predicted base poses and measuring the fraction of poses that lead to successful manipulation in the simulator. We report Top1 and Top5.

For MSR, we rank floor points by predicted affordance, select the Top-k unique base poses, and execute the simulator-defined attempt from each pose on the unseen split. **Submission format.** Predictions are per-episode JSON with per-floor-point scores aligned to provided in-

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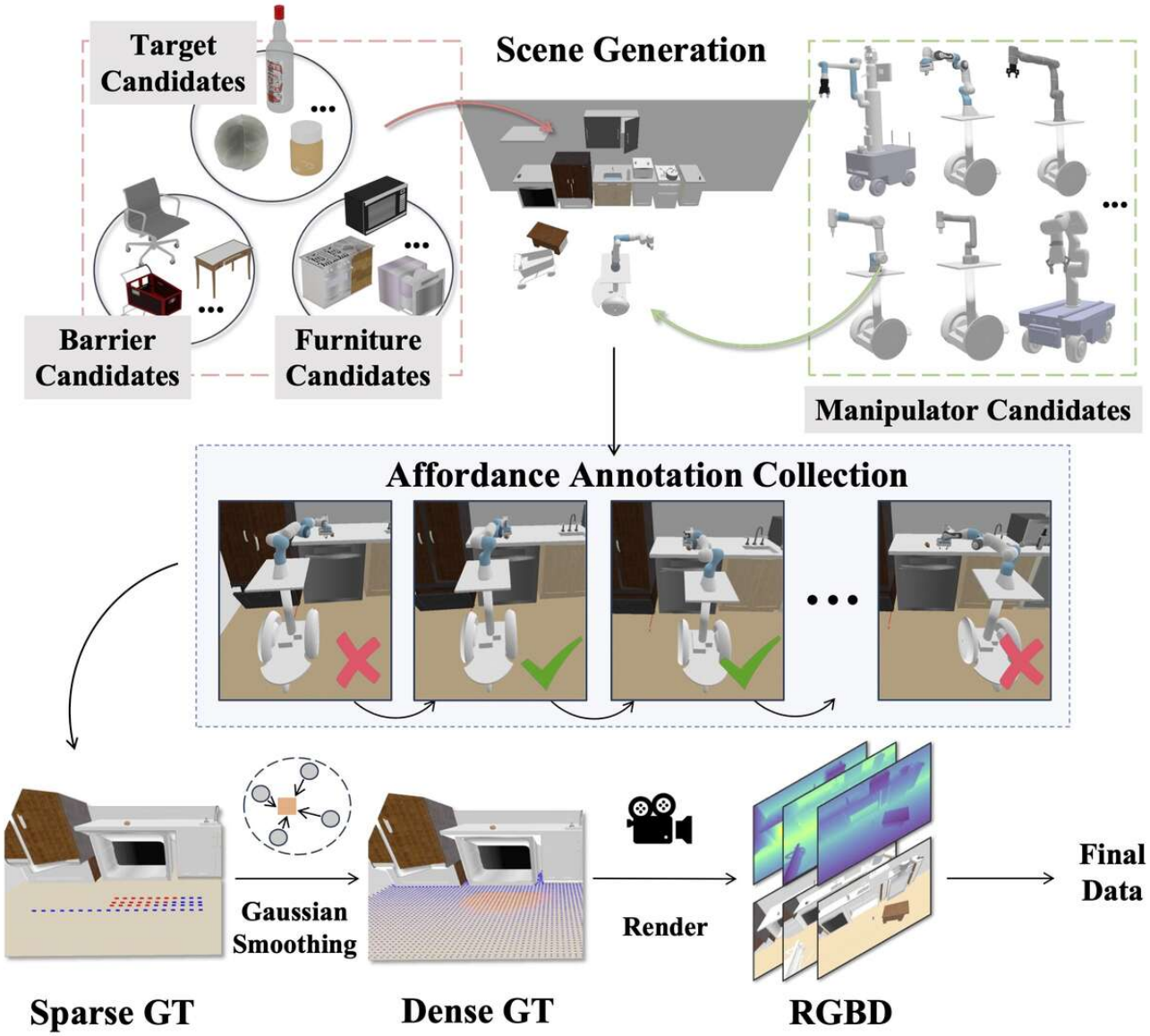

Figure 2. **Benchmark pipeline.** Scene synthesis, robot sampling, viewpoint selection, manipulation-grounded labeling, frame alignment, and densification produce continuous floor affordance maps aligned to first-person RGB-D and point clouds.

dices; our checker validates indices, shapes, and NaN-free outputs for strict comparability.

Reproducibility. All metrics are computed on the held-out unseen-scenes split with fixed RGB-D, point clouds, and robot parameters. The pipeline is deterministic; there is no human annotation and thus no inter-annotator variability. **Statistical reporting.** We recommend reporting 95% confidence intervals via nonparametric bootstrap over episodes for PCC and SIM; the released scripts include this option for transparent uncertainty. To reduce ranking volatility, we use paired bootstrap resamples and report PCC/SIM with 10k resamples. For MSR, ties in Top-k scores are broken deterministically by episode and index order to ensure exact reproducibility across submissions.

Fair-comparison guardrails. The evaluator enforces that submissions are single-pass functions of the provided inputs (no retrieval from private maps or cross-episode state on the test set); per-episode parameter adaptation and test-time optimization are disallowed; and fixed discretization, NMS, and Top-k policies are applied uniformly within the evaluator. Inputs are hashed and predictions are validated for shape, NaN-free values, and non-degenerate per-episode

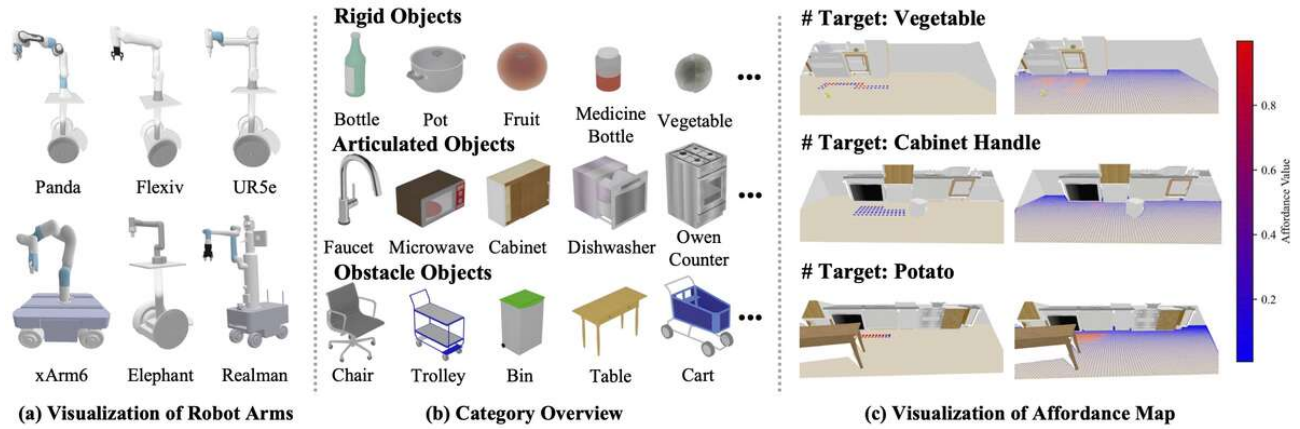

Figure 3. **Coverage and distributions.** Scene and target diversity across layouts, object categories, base–target distances, occlusion, and clutter. The benchmark spans 569 kitchens and 127k episodes, enabling slice analyses by reach, base height, and visibility.

variance. We release a small public probe split for hyperparameter selection; all reported scores must come from the frozen unseen split. Alongside accuracy, we require reporting mean per-episode latency on a reference GPU to contextualize efficiency.

5. Reference Baselines

We evaluate adapted 3D perception baselines—PointNet++ [5], VoteNet [6], and H3DNet [9]—for dense affordance regression from point clouds. We also provide a lightweight reference model, NavAff, to anchor the task.

NavAff (reference). NavAff fuses global and floor-level 3D point features with robot morphology and predicts a dense affordance over the floor cloud. Concretely, two PointNet++-style encoders [5] process a global point cloud with a target/obstacle indicator and a floor point cloud, respectively. A small MLP encodes robot height and reach, and a cross-attention module aligns global context with floor features. A point-based decoder with feature propagation outputs a per-point feasibility score in $[0, 1]$. The architecture is deliberately simple to facilitate replication.

6. Experiments

Setup. We train on 456 scenes (102,687 episodes) and evaluate on 113 unseen scenes (24,656 episodes). Inputs include the aligned floor and global point clouds and robot parameters. We train with mean squared error on valid floor points and report all metrics on the unseen split. Implementation details follow the released code.

Reproducibility. We fix seeds; splits and simulator configs are versioned; released scripts regenerate the exact train/val/test sets and metrics. Evaluation is deterministic and directly comparable.

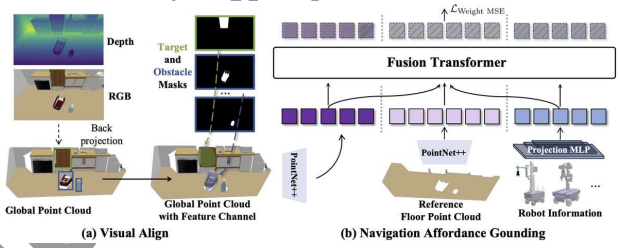

Figure 4. **NavAff.** Lightweight reference that fuses global and floor point features with robot morphology to predict dense floor affordances.

Main results. Table 2 summarizes quantitative performance. Adapted 3D detectors underperform NavAff on correlation- and similarity-based metrics, suggesting that features and fusion geared toward manipulability are beneficial.

Table 2. **Main results.** Quantitative evaluation of navigation affordance grounding performance of NavAff.

Method	RMSE ↓	logMSE ↓	PCC ↑	SIM ↑
PointNet++ [5]	0.164	0.0142	0.565	0.589
VoteNet [6]	0.167	0.0143	0.543	0.570
H3DNet [9]	0.174	0.0156	0.503	0.522
NavAff	**0.147**	**0.0115**	**0.680**	**0.696**

End-to-end utility. We evaluate MSR by executing manipulation from the Top-k predicted base poses. NavAff improves Top1/Top5 MSR over baselines (Table 3).

Ablations. Removing visuomotor alignment (VAM) or global context degrades PCC and SIM substantially, while removing robot information has a smaller effect, indicating that 3D scene context and feature alignment are most criti-

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Table 3. **Manipulation Success Rate (MSR)**.

MSR	Random	H3DNet	VoteNet	PointNet++	NavAff
`Top1`	0.080	0.54	0.56	0.60	**0.72**
`Top5`	0.046	0.47	0.53	0.58	**0.66**

Table 4. **Ablation study results of NavAff.**

	RMSE ↓	logMSE ↓	PCC ↑	SIM ↑	`top1`MSR ↑
NavAff	0.147	0.0115	0.680	0.696	0.72
w/o robot information	0.148	0.0115	0.670	0.688	0.70
w/o VAM	0.165	0.0140	0.562	0.589	0.63
w/o global point cloud	0.168	0.0144	0.534	0.568	0.58

cal in this setup (Table 4).

Qualitative results and sim-to-real. Figure 5 illustrates qualitative affordance predictions versus ground truth across methods; Figure 6 shows a pipeline that applies a simulator-trained model to real kitchens with commodity RGB-D, yielding plausible floor affordances.

7. Evaluation Protocol

We standardize inputs, predictions, and reporting. Systems receive RGB-D and robot parameters and must output a continuous scalar field over a local floor patch in the agent frame. We evaluate with RMSE and logMSE in the field domain and correlation-based metrics (PCC and cosine SIM) to assess shape fidelity. For decision-centric evaluation, we extract Top-k base poses by non-maximum suppression on the predicted field with a minimum spatial separation; MSR@k executes a short manipulation attempt from each pose and counts success if the target is reached without collision within limits. Rankings are by primary PCC and reported alongside RMSE/logMSE and MSR@1/5. To ensure comparability, we fix train/val/test splits, random seeds for any sampling, and exact discretization of the evaluation patch. Cross-robot evaluation is supported by swapping morphology parameters while keeping scenes fixed, enabling like-for-like comparisons across embodiments.

Reproducibility and release. We provide code, configs, and deterministic seeds for data generation, splits, and evaluation. Each episode includes a content hash over scene assets, articulation states, robot parameters, and camera pose. The evaluation script enforces identical discretization, interpolation, and NMS thresholds and emits a manifest recording code version, CUDA, and driver info. To support fair comparison, we release reference Docker images and precomputed features (RGB, depth, and point clouds) for the test split, plus unit tests that check metric invariances under small depth noise and rigid transforms.

8. Validation and Quality Checks

We verify that MoMa-Kitchen provides meaningful supervision and is not trivially solvable:

- **Task validity:** physical attempts directly supervise feasibility and reject occluded or colliding poses. - **Distributional diversity:** scenes are procedurally varied with clutter and a broad set of rigid and articulated objects, inducing diverse reach and collision constraints. - **Difficulty:** baseline detectors adapted from object detection underperform a simple affordance-focused reference, indicating non-triviality and the need for tailored fusion. - **Robustness:** metrics agree in trend across RMSE, PCC, and SIM; improvements in PCC correlate with MSR. The pipeline is deterministic and reproducible. - **Known limitations:** supervision relies on simulated grasp/suction and floor-level base placement in kitchens; dynamic obstacles and non-kitchen domains are out of scope. We discuss these in §10.

Beyond these checks, we hash scene and episode manifests to rule out split leakage. Under small depth noise and pose jitter, system rankings are stable; RMSE shifts modestly while PCC and SIM remain effectively unchanged. Replacing PCC with Kendall tau-b preserves pairwise ordering for most model pairs, and varying the floor sampling resolution by $\pm 25\%$ yields the same winner across all metrics, indicating robustness to metric and discretization choices.

9. Analysis

Standard slices. To support consistent analysis, we release per-episode slice definitions: occlusion quartiles (by target visibility), distance bins (base–target geodesic), reach ratio (target height/arm reach), articulation state (articulated vs rigid), and corridor width (minimum free space around the base). **Slice diagnostics.** We analyze performance by ablations: removing global point clouds or VAM substantially reduces PCC (from 0.680 to 0.534 and 0.562 respectively), highlighting the importance of global context and cross-modal alignment for affordance prediction. Robot-parameter removal has a smaller impact (PCC 0.670), suggesting that in many kitchen configurations, scene geometry dominates feasibility, though morphology still informs fine-grained reach.

Ranking stability and MSR. Differences that are moderate in RMSE are amplified in MSR, indicating that high-correlation predictions more reliably surface feasible base poses in the Top-k set—a desirable property for planners that sample a few candidates.

What the benchmark measures. MoMa-Kitchen measures the ability to anticipate manipulation feasibility from a single egocentric view and robot parameters. It does not evaluate the full navigation stack, grasp planning beyond simple end-effectors, or long-horizon task execution; rather, it isolates the base-placement decision at execution time.

Cross-scene variability and error modes. Errors concentrate in occluded layouts and narrow passages (false positives) and near base-orientation boundaries; large-swing ar-

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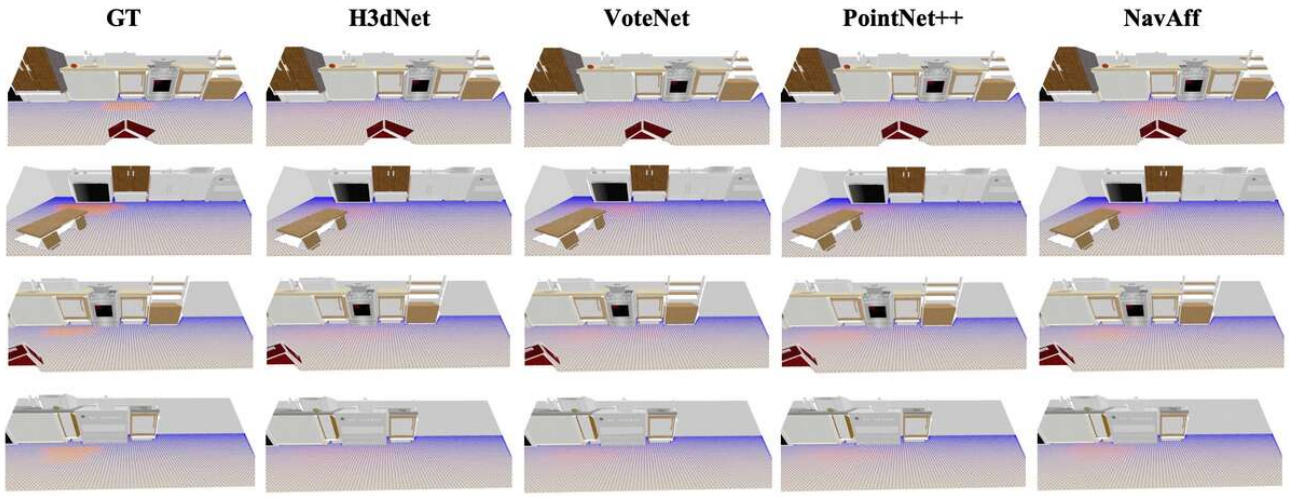

Figure 5. **Qualitative comparison** of predicted floor affordances across methods versus ground truth.

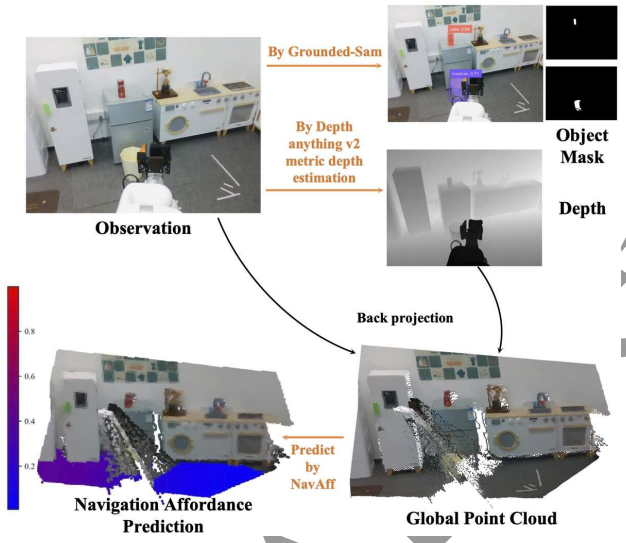

Figure 6. **Sim-to-real.** Pipeline applying a simulator-trained model to real kitchens with RGB-D and open-vocabulary segmentation.

ticulations induce conservatism that reduces low-k MSR. Calibration analyses show monotonic increases in MSR across predicted score quantiles, and simple temperature scaling improves Top5 MSR by a small but consistent margin without altering rankings. **Expert alignment.** We include an optional protocol for human sanity checks: experts inspect egocentric RGB-D and predicted fields to label feasible base poses with a standardized rubric. The release provides forms and scripts to compute agreement with MSR and to surface systematic disagreements for audit.

Cross-embodiment generalization. Beyond single-robot evaluation, we test whether a trained predictor preserves ordering across morphologies by swapping base footprint and arm reach while keeping scenes and cameras fixed. To decouple geometry-limited headroom, we normalize MSR by an oracle that has access to the ground-truth field. We find that most models retain coarse ordering, but ranking inversions appear when the base footprint shrinks or the arm reach increases, suggesting that some predictors entangle scene geometry with morphology scale. Reporting normalized MSR alongside PCC/SIM across two or more embodiments therefore provides a more faithful picture of morphological reasoning and guards against overfitting to a fixed footprint.

10. Discussion

Use and scope. We recommend using MoMa-Kitchen to benchmark perception-and-planning modules that select base poses conditioned on robot morphology. The MSR protocol facilitates evaluating how prediction quality translates to downstream utility. MoMa-Kitchen complements existing navigation datasets by incorporating manipulability into evaluation.

Intended comparisons and misuse. Because MoMa-Kitchen isolates base-placement feasibility, comparisons should hold planners, grasping heuristics, and discretization fixed, varying only the perception module or morphology of interest. Reporting MSR alongside PCC/SIM aids interpretation: large MSR deltas with negligible correlation change often indicate overfitting to threshold artifacts or sampling effects. Cross-embodiment comparisons should control for sensor pose and field of view; swapping only morphology while keeping scenes and cameras fixed yields most informative deltas. We discourage aggregating

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scores across scenes with different target types unless slice-weighting is reported, and we recommend publishing Top-k success curves (not just a single k) to reveal sensitivity. Finally, when using MoMa-Kitchen for sim-to-real studies, release calibration procedures and any depth filtering used, as these can materially affect absolute MSR without reflecting model capability.

Benchmark reporting and reproducibility. To reduce variance and enable fair comparison, we ask participants to report training compute (GPU hours), parameter count, and per-episode inference latency; the evaluator logs end-to-end wall time and enforces identical discretization and NMS thresholds. We recommend paired bootstrap confidence intervals for PCC/SIM and McNemar tests on MSR decisions when comparing two systems. To limit overfitting, the public leaderboard is refreshed periodically with a small blind extension and we cap test submissions per team per week. We release both a frozen test split and a development split; only the former supports claims, and ablations should use the latter.

Responsible use. Use MoMa-Kitchen for diagnostic analysis rather than leaderboard chasing. Report per-slice scores and calibrated error maps, and do not tune or distill on the frozen test split. Interpret claims jointly with the primary metric and MSR. **Limitations.** Coverage is limited to kitchen scenes and floor-level base placement; articulated interactions are supervised through simple grasps/suction rather than full manipulation policies; and sim-to-real results are qualitative. While the pipeline is contamination-resistant and reproducible, it may contain residual artifacts from asset geometry and simulator physics. We therefore discourage interpreting high scores as general mobile manipulation ability beyond the defined scope.

11. Conclusion

We present MoMa-Kitchen, a large-scale benchmark that reframes last-mile navigation as robot-conditioned floor affordance prediction. A fully automated, physics-grounded pipeline produces dense, continuous supervision aligned to egocentric RGB-D and 3D geometry. Experiments with strong 3D backbones and a lightweight reference model reveal that affordance-grounded evaluation is challenging yet informative, with clear links to downstream success via MSR. We hope MoMa-Kitchen will catalyze research on base placement policies and perception models that bridge navigation and manipulation in cluttered homes.

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